Khalifeh Basiri

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Education

Carleton University — Ottawa, ON
Bachelor of Computer Science, GPA: 3.5/4.0 — *2026*

Work Experience

Fisheries and Oceans Canada (DFO) | Government of Canada — Ottawa, ON
Software Developer (internship) — *Aug. 2024 – Dec. 2025*

- Built a bilingual Python geographic-name validator using the GNBC REST API and geospatial matching, replacing manual entry-by-entry file checks with validation completed in minutes.
- Expanded the validator into a modular Name Manager with comparison, GNBC search, and geospatial helper workflows; CHS Atlantic reported one process fell from two weeks to a few hours.
- Led a three-person Agile team and worked directly with DFO clients through requirements, design, development, demos, deployment, and user training for the Name Manager.
- Performed integration, regression, and user-acceptance testing (UAT) across API, database, threading, and geospatial workflows; validated releases with users in five DFO regions and the Canadian Coast Guard and supported production.

Programmer (internship) — *Sep. 2023 – Aug. 2024*

- Engineered a multithreaded Python Storage Explorer using Azure Blob Inventory, MongoDB, and SQLite to search and export metadata across a 420 TB+ Azure storage estate; cut DFO staff's file-location searches from hours to seconds or minutes.
- Implemented metadata refresh handling that kept searches responsive and preserved the last valid database after failed updates, allowing staff to continue using the Explorer.

Floxy — Remote
Frontend Developer — *Mar. 2025 – May 2025*

- Built Floxy's responsive Next.js site with React, TypeScript, and Tailwind CSS for a platform with 20,000+ customers; the site reached 23.9K monthly visits within four months.
- Organized the site into reusable components and centralized content models, keeping product pages and language-specific API examples consistent and easy to update.

Projects

ZDash | *TypeScript, React, Electron, SQLite, OpenAI API, LangChain, LangGraph* — Product Site

- Engineered a Windows vehicle diagnostics app for Nissan and Infiniti CONSULT-I ECUs with live telemetry, fault-code diagnostics, and SQLite/WAL session recording.
- Built a diagnostic agent using LangChain tools and LangGraph orchestration, with scoped tool permissions and SQLite checkpoints for resumable investigations; prototyped RAG over 8,618 service-manual pages using PostgreSQL/pgvector hybrid search and page-level citations.

Atelier – Art Community | *Angular, Express, PostgreSQL, MongoDB, Redis* — GitHub | Live Demo

- Built and deployed an Angular/Express art community with PostgreSQL relationships, MongoDB artwork storage, Redis caching, and signed cursor pagination; handled interrupted publication through reconciliation and compensating writes.
- Implemented rotating refresh tokens, replay detection, CSRF protection, and session revocation; built a GitHub Actions CI/CD pipeline with automated security testing and staged deployments.

Job-Tracker-AI | *Python, PySide6, OpenAI API, SQLite, Pydantic* — GitHub | Download

- Built a Python desktop job tracker with Gmail/Outlook OAuth integration and a two-stage AI pipeline using the OpenAI Decisions API for email classification and the Responses API for structured extraction.

Tandem t:slim X2 Insulin Pump Simulator | *C++, Qt, SQLite, UML* — GitHub

- Designed bolus calculations and CGM-driven insulin-control logic in a four-person C++/Qt team, adding safety limits, suspend/resume behavior, alerts, and SQLite-backed event history.

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C++, C, SQL, HTML, CSS
Frameworks and libraries: React, Next.js, Angular, Express, Node.js, Electron, Flask, Qt, Qt Charts, TensorFlow, TensorFlow Lite, MediaPipe, Tailwind CSS, pytest, LangChain, LangGraph
Cloud, APIs, and tools: Microsoft Azure, Azure Blob Storage, QGIS, Docker, Git, Linux, REST APIs, OpenAI API, OAuth
Databases and caching: PostgreSQL, pgvector, MongoDB, SQLite, Redis, Oracle, Convex